

Exercise 2: Computational Aeroacoustic Simulation

Description

This problem considers a 2D cylindrical obstacle (diameter $d=0.02$ m) within an incompressible flow. Inflow velocity $U_0 = 10$ m/s, kinematic viscosity $\nu = 0.001$ m²/s, Reynolds number $Re \approx 200$.

Air at 20°C should be defined as material for the simulation (Material parameters are inside the `air.xml` file.). The speed of sound of air can be assumed to be $c \approx 343.5 \frac{m}{s}$. The computational aeroacoustic domain is depicted below.

Inside the source region, a transient incompressible CFD Simulation is performed. Thereafter, aeroacoustic source terms can be computed from the obtained flow field. We then solve the Perturbed Convective Wave Equation (PCWE) assuming a low Mach number ($Ma < 0.3$). The convective wave equation fully describes acoustic sound generated by incompressible flow structures and their wave propagation. The only unknown is the acoustic scalar potential ψ^a . In order to receive the acoustic pressure, we have to derive the scalar potential and scale it with the density.

Tasks

1. Perform an openFoam Simulation of the Cylinder (monitor the transient behavior from startup until the vortex street is in a steady state).
 - Plot flow pressure p and velocity $|U|$ at the monitoring points: MP1 (1,0,0), MP2 (-1,0,0), MP3 (0,0.01,0), MP4 (0,2.9,0) **(12 Points)**
 - Calculate the Strouhal number from the pressure data. **(4 Points)**
 2. Write down the data from the Flow simulation into an Enight file for further processing (look at the PCWE and neglect convective sources), and write the necessary flow quantities to the storage. **(2 Points)**
 3. Calculate the aeroacoustic source and interpolate the sources conservatively. **(4 Points)**
 4. Create input-xml files for the required openCFS simulations to perform a transient simulation of the computed source field. **(8 Points)**
 5. Document the results you got from the openCFS simulation by answering the questions below.
 - Check the decay in pressure amplitude and state whether the microphones and the microphone array are positioned in the acoustic near- or far-field. **(4 Points)**
 - Calculate and plot the SPL in dB vs. frequency in Hz (for mic1 and mic2). You can use the resulting `.hist` files. **(4 Points)**
 - Add a directivity plot of the acoustic pressure, radius 2 m, and frequency 95 and 190 Hz. You can use the resulting files in the folder mics. **(4 Points)**
 - Implement a postprocessing script to calculate a directivity plot of the RMS (root mean squared) of the acoustic intensity, radius 2 m. You can use the resulting files in the folder mics. **(4 Points)**
 - Discuss the results **(4 Points)**
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Hints

- Use the provided template simulation setup. (Can be downloaded [here](#))
- Use the provided xml templates `interpolation.xml`, `propagation.xml` and `air.xml`
- Use the provided mesh files: `sourceRegion.cfs`, `propagationRegion.cfs` and `PML.cfs`
- The evaluation of microphone positions defined in `MicPos.txt` can be found in `evaluate_mics.py`
- Use the provided file `evaluate_hist.py`.
- State the reference pressure for any SPL in dB.
- Additional information, application examples, etc. can be found at: <https://opencfs.gitlab.io/userdocu/>

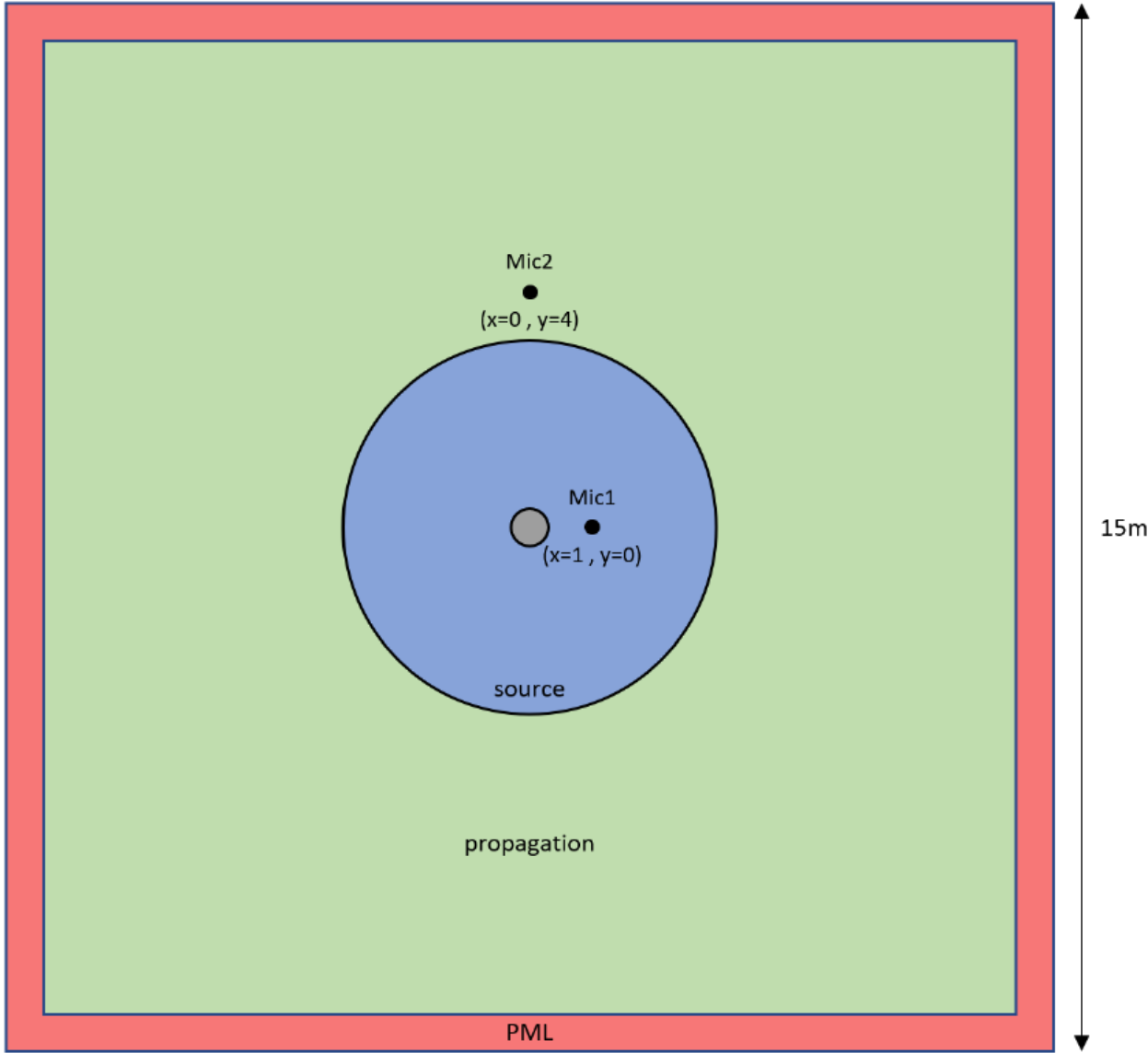


Figure 1: Simulation Domain

Submission

Submit all your input and postprocessing files (*.xml, *.py) as well as a concise PDF report of your results. Name the archive according to the format `HW1_eNNNNNNN_Lastname.zip`.

Download and install Virtual Machine

For this example, a Debian Linux VM is provided, which includes all software required for the example. To run the virtual machine without problems, the host machine should have more than 4 GB RAM and 25 GB free disk space.

- Download and install *Oracle VirtualBox* and *VirtualBox Extension Pack* for your host platform (<https://www.virtualbox.org/wiki/Downloads>).
- Download the Virtual Machine (<https://cloud.tugraz.at/index.php/s/5wqgSyLEWSaDY9M>).
- Open VirtualBox and select *File > Import Appliance* to import the downloaded virtual machine
- In the *Application Setting* you can adjust, e.g. the RAM and the *Machine Base Folder* to specify the location, where you have stored the VM.
- Click *Import*
- Double click on the Virtual Machine to start it.
- Standard user is *user* and has password *user*.
- To change the password enter `$ passwd` in a terminal.

Tips for using the Linux Terminal

- Do not use spaces in file names
- Use the **TAB** key to auto-complete commands
- Use the **ArrowUp** key to show the last command
- Type something and press the **PageUp** key to show the last command starting with what you typed
- Do not close the terminal window if you started GUI programs from it (this will kill the program)
- Press **Ctrl + C** to cancel a running command.
- Copy-and-Paste in a Linux Terminal works by selecting text to copy and pressing the middle mouse button to paste.

Common shell commands:

command	description
<code>pwd</code>	print working directory
<code>ls</code>	list files
<code>cat file</code>	print file <i>file</i> to screen
<code>grep</code>	print lines matching a pattern
<code>cd dest</code>	change directory to <i>dest</i>
<code>cp source dest</code>	copy file <i>source</i> to <i>dest</i>
<code>mv source dest</code>	move <i>source</i> to <i>dest</i> (rename)
<code>mkdir directory</code>	make directory <i>directory</i>
<code>man command</code>	show manual page of command <i>command</i>
<code>command -h</code>	show help to <i>command</i>
<code>which command</code>	show binary for <i>command</i> in PATH

Perform Transient Flow Simulation with openFOAM

The flow simulation requires considerable computation power, thus configuring the virtual machine to access 4 cores/threads and ≥ 4 GB RAM is recommended

- Extract the simulation setup in your working directory (root folder of the template directory is referred to as `~` in the following)
 - It is important, that your template folder is copied somewhere locally (e.g. `/home/USERNAME/template`) and not inside a shared folder!
- Open a terminal and navigate to `~/1_cfd` directory
- Start the openFoam environment with the command `$ openfoam8-linux`
- Once you are inside the virtual environment for the first time, type `$ sh PrepareOpenFoamContainer.sh` to install missing python packages (*this has to be done just once*)
- Open the simulation folder: `$ cd cfd`
- Run the flow simulation with the initial setup using `$ icoFoam`. Alternatively, we can save time by running the simulation in parallel:
 1. Split domain into 4 parts using `$ decomposePar`
 2. Run the simulation in parallel as a background process
 - `$ mpirun -np 4 icoFoam -parallel > log &`
 3. You can check the computation status by viewing the end of the log file with `tail -n 20 log`
 4. Combine the results using `$ reconstructPar -newTimes`
- View the flow field in ParaView using `$ paraFoam &`
 - Plot flow pressure p and velocity U at the monitoring points: MP1 (1,0,0), MP2 (-1,0,0), MP3 (0,0.01,0), MP4 (0,2.9,0)
 1. Find data (Hotkey V)
 2. Search for the closest point to the monitoring point
 3. Plot Selection over Time
 - Velocity and pressure fields should show a steady oscillation at the end of the simulated timeframe indicating transient phenomena of the initialization of the simulation have vanished. However, the data resolution in time is not sufficient to resolve the oscillations of the vortex street.
- Export source data
 - Edit `~/1_cfd/cfd/system/controlDict` with a text editor eg. type `$ gedit system/controlDict`
 - We want to simulate the acoustics for 0.2s, so we need to change `startTime` to 0.5 and `endTime` to 0.7.
 - To adequately resolve the oscillations save every 20th timestep of the CFD simulation. Therefore, change `writeInterval` to 20.
 - Run the simulation as before
 - Click *Refresh Times* or reopen ParaView with `$ paraFoam` command and view the flow field
 - Export Pressure data in Ensight-Case file format using
 - * `$ foamToEnight -fields '( "p.*" )' -time 0.5:0.7`
 - You can view the exported data in ParaView by opening `~/1_cfd/cfd/EnSight/cfd.case`
- Exit the openFoam environment with the command `$ exit`

Interpolate the Flow Field and Compute Acoustic Sources

The VM has Eclipse IDE installed as an XML Editor to edit the openCFS Setup files. You can launch Eclipse with `$ eclipse file.xml`

- Navigate to `~/2_sourceInterpolation` and open `interpolation.xml` with your xml editor
- Define the `cfd.case` in your input filter and the target mesh `~/0_mesh/sourceRegion.h5` in the interpolation filters
- Calculate interpolated acoustic sources: `$ cfsdat -tX interpolation` using X threads in parallel.
- View the acoustic sources in ParaView by opening
 - `~/2_sourceInterpolation/results_hdf5/source_acouRhsLoad.cfs`

Acoustic Simulation Setup

- Define input files
- Include material file (`air.xml`)
- Assign material to regions (air 20 degrees)

- Define interfaces between the different regions/parts (Nitsche-type Mortaring)
- Define microphone points for evaluation
- Run the acoustic propagation simulation: `$ cfs -tX propagation` using X threads in parallel.

Post Processing

You can edit the template python scripts with `$ gedit file.py` and run them with `$ python3 file.py` in the VM. You can also edit and run your postprocessing scripts locally on your host platform (requires Python 3 with `numpy` and `matplotlib` packages installed). If you prefer postprocessing with Matlab you are free to write your own postprocessing script.

- View the field result in ParaView and evaluate the amplitude decay of the acoustic pressure field. Discuss whether each evaluation point lies in the acoustic near- or far-field.
- Use the provided python script `evaluate_hist.py` to read the `.hist` files and calculate the amplitude spectrum (using Welch's method).
- Calculate the SPL from the amplitude spectrum and plot it over the frequency.
- Use the provided python script `evaluate_mics.py` to read the microphone results and calculate the Fourier transform.
- Generate the directivity plot of the Fourier transformed acoustic pressure
- Write your own postprocessing script to create a directivity plot of the RMS of the acoustic intensity.